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Azad National Institute of Technology Bhopal, Bhopal – 462003 (An

Institution of National Importance under Ministry of Education, Govt. of India)

Department of Computer Science & Engineering

B.Tech. VI Semester (EE1)

Data Structures (CSE352)

Assignment-III

1. What are the time complexity of the following operations for a stack & queue ?
 - a) Insertion
 - b) Deletion
 - c) Searching
2. An initial queue is represented as: [A, B, C, D, E]
Perform the following operations:
 - a. Enqueue "F"
 - b. Dequeue two elements
 - c. Enqueue "G" and "H"
 - d. Dequeue one element
3. Consider the following scenarios. For each, state whether you would prefer to use a stack or a queue and justify your choice:
 - a) Managing ticket booking requests in a booking counter.
 - b) Implementing an undo-redo functionality in a text editor
4. An initial stack is represented as: [1, 2, 3, 4, 5] .
Perform these operations:
 - a. Push 6 and 7
 - b. Pop three elements
 - c. Push 8
 - d. Pop one element
5. A stack initially contains N elements. If X push operations and Y pop operations are performed, derive the formula to determine the total number of elements left in the stack. For example,
Input: N = 10, X = 5, Y = 4 Output: 11 (11 elements are left after 5 push and 4 pop operations are performed assuming the stack initially contains 10 elements).